

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2008

2134212

MATHEMATICS

READ THESE INSTRUCTION CAREFULLY

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question.
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer. Mark it on the **ANSWER SHEET**, not in this question booklet.

HOW TO USE THE ANSWER SHEET

4. Use only ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER**YOUR NAME****NAME OF YOUR SCHOOL**

6. By drawing a dark line inside the correct numbered boxes mark your full Index (i.e. School Code Number and the three-figured Candidate's Number) in the grid near the top of the answer sheet.
7. Do not make any mark outside the boxes.
8. Keep the answer sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answer are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

11. What is the value of $\frac{6(24 - 18) + 6 \times 4}{6}$?

6

- A. 30
- B. 25
- C. 10
- D. 28

1 [A] [B] [C] [D] 11 [A] [B] [C] [D] 21 [A] [B] [C] [D] 31 [A] [B] [C] [D]

In the set, the box with the letter D printed in it marked.

11. Your dark line MUST be within the box.
12. For each question ONLY ONE box is to be marked in each set of the four boxes.

The Kenya National Examination Council, 2008.

K.C.P.E MATHEMATICS 2008

- Which one of the following numbers is one million five hundred three hundred and eight?
 - 1005308
 - 1050308
 - 1500308
 - 1005038
- What is the value of $\frac{6^2 - 6}{3} + 3$?
 - 5
 - 11
 - 13
 - 31
- What is the number 21.046 rounded off to 2 decimal places?
 - 21.46
 - 21.04
 - 21.00
 - 21.05
- What is the total value of digit 6 in the number 40607580?
 - Six million
 - Six hundred thousand
 - Six thousand
 - Hundred thousand
- The table below shows the number of 90kg bags of maize that Mutiso sold to a miller in the years 1998 to 2002.

Year	1998	1999	2000	2002
No of bags	13,092	14, 290	13,995	13,226

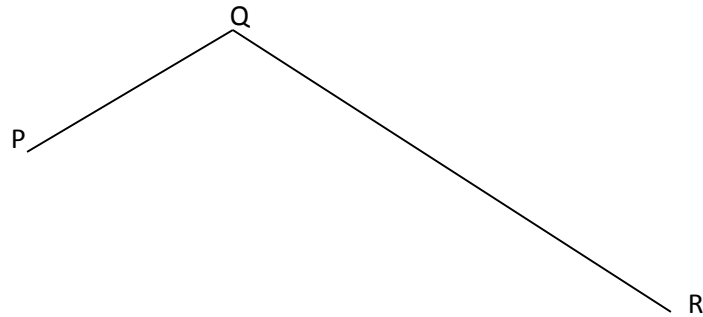
In which year did Mutiso have the least increase in the number of bags sold?

- Year 2001 and 2002
 - Years 2000 and 2001
 - Years 1999 and 2000
 - Years 1998 and 1999
- In the figure below, PS is parallel to VU, and $\angle TRS = 75^\circ$ and $\angle TUV = 130^\circ$

What is the size of angle RTQ?

- 25°
 - 55°
 - 50°
 - 105°
- What is the GREATEST Common Divisor of 120 and 270?
 - 3

- B. 10
C. 20
D. 1080
8. What is the capacity of the container which measures 1.5m by 1.0m by 0.5m in litres?
A. 7 500
B. 750
C. 75
D. 7.5
9. On the figure below, draw perpendicular bisector of lines PQ and QR to meet at point X.



What is the length of QX?

- A. 5.2cm
B. 4.4cm
C. 7.0cm
D. 5.5cm
10. What is the next number in the pattern
1, 4, 10, 19, ____?
A. 28
B. 29
C. 31
D. 34
11. Atieno had the following amount in her saving box. How much money did she gave altogether?

Value of coins in shillings	1	5	10	20	50	100	200	500	1000
Numbers of coins/notes	25	10	25	20	5	6	4	1	1

- A. SH. 97
 - B. Sh. 3 875
 - C. Sh. 1 983
 - D. Sh. 1 886
12. A saleslady earns a salary of sh. 25 000 per month. She also gets 4% commission of goods sold above a total of 150 000. In one month she sold goods worth sh. 350 000. How much money altogether did she earn that month?
- A. Sh. 8 000
 - B. Sh 39 000
 - C. Sh.b31 000
 - D. Sh. 33 000
13. One the line MN drawn below, complete the quadrilateral MNPQ in which lines NP = 3cm and line QM =7cm. Angle QMN =60° and angle MNP = 90°

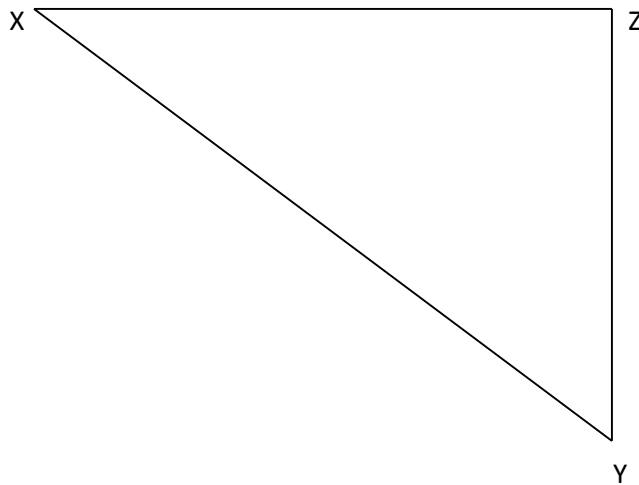


- What is the length of diagonal MP?
- A. 3.4cm
 - B. 6.2cm
 - C. 5.0cm
 - D. 5.8cm
14. A shopkeeper spent sh. 880 to buy 16 plates and then sold them making a profit of 20%. For how much did the shopkeeper sell each plate?
- A. Sh. 44
 - B. Sh. 55
 - C. Sh. 66
 - D. Sh. 1 056
15. In a trapezium the two parallel sides are 6cm and 8cm long. The perpendicular distance between the parallel sides is 4cm. What is the area of the trapezium?
- A. 24cm²
 - B. 28cm²
 - C. 32cm²

- D. 56cm^2
16. A cyclist left home for town, a distance of 30km, at 7.00a.m. After travelling for 30 minutes at a speed of 10km/h, The bicycle got a puncture which took 15 minutes to repair. The cyclist then resumed the journey and reached the town at 9.00a.m. which one of the following graphs below represents the cyclists journey?
17. At a film show, $\frac{1}{5}$ of the attendants were boys while $\frac{1}{4}$ were girls. Two fifth of the remainder were men and the rest were women. What fraction of the total attendants were women?
- A. $\frac{11}{20}$
 B. $\frac{11}{50}$
 C. $\frac{3}{20}$
 D. $\frac{33}{100}$
18. The mean of 7 numbers is 4, Six of the numbers are 2,3,1,2,7,8. What is the mean of the 7 numbers?
- A. 2
 B. 2.5
 C. 3
 D. 5
19. What is the value of $\frac{8 + 0.5 \times 0.3 - 0.7}{0.09}$?
- A. 9
 B. 90
 C. 27.7
 D. 23.61
20. The scale on the map is 1 : 200. What length on the map would represent a distance of 50 metres?
- A. 0.25
 B. 25
 C. 4cm
 D. 2.5cm
21. Otieno, Leila , Rotich and Furaha shared sh. 840. Otieno got twice as much as Leila got three times As much as Rotiech while Rotiech got half of what Furaha got. What was the difference between Otieno's share and Furaha's share?
- A. Sh. 350
 B. Sh. 420
 C. Sh. 210
 D. Sh. 280
22. Mwasi borrowed sh. 50 000 from a bank that charged compound interest at rate of 25% p.a. How much should he pay the bank at the end of two years?

- A. Sh. 78 125
- B. Sh. 75 000
- C. Sh. 62 500
- D. Sh. 28 125

23. Draw the circle that passes through the points X,Y and Z of the triangle below.



What is the measure of the radius of the circle?

- A. 2.0cm
- B. 4.3cm
- C. 5.0cm
- D. 3.0cm

24. What is the value of $\frac{2q-r}{y} + n$

Where $n=5$, $y=2n$, $q=n+9$ and $r=q-6$?

- A. 7
- B. $2\frac{1}{2}$
- C. $6\frac{1}{5}$
- D. 11

25. An open cylindrical tin has a diameter of 14cm and a height of 10cm. What is the surface area of the tin in cm^2 ? (Take $\pi=22/7$)

- A. 1 540
- B. 1 496
- C. 748
- D. 594

26. How many faces (F), vertices (V) and edges (E) does a triangular prism have?

- | F | V | E |
|----------|----------|----------|
| A. 4 | 4 | 6 |

- B. 6 8 12
C. 5 5 8
D. 5 6 9
27. Ahmed earns sh. 560 500 per year. Oloo earns three times as much money as Alice earns twice as much as Ahmed . How much money, altogether , do they all earn in a year?
- A. Sh. 1 121 000
B. Sh. 3 363 000
C. Sh. 4 484 000
D. Sh. 5 044 500
28. Musa left home at 11.45 am and took 1h 20min to travel to town. After staying in town for one hour , he then travelled back home. The time taken to travel to town was twice that taken for the return journey. At what time did he reached home?
- A. 3.25p.m
B. 2.45p.m
C. 4. 45p.m
D. 1.45p.m
29. A kiosk sold soda, juice, porridge and tea. A pie chart was drawn to represent the number of people who took each drink. Those who took juice were represented by 140° and porridge by 40° . The size of angle of those who took tea was twice the angle for soda. Twenty people took porridge. How many more people took juice than those who took tea?
- A. 10
B. 60
C. 70
D. 130
30. In the figure below, GFE and GHJ are straight lines. Line GF = FH = HJ and $\angle FGH = 50^\circ$.

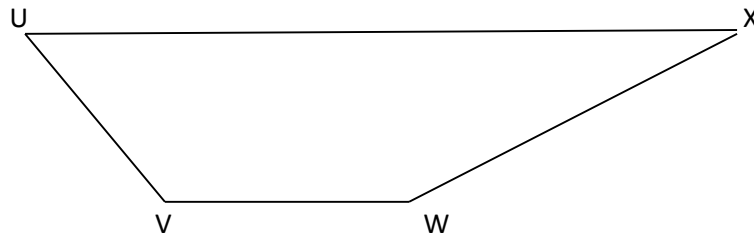
What is the measures of angle EFJ?

- A. 130°
B. 75°
C. 80°
D. 105°
31. Which one of the following expression is the simplest form of $\frac{X + 3(x + 2y) - 2x + 2y}{X}$?
- A. $\frac{2x + 4y}{X}$
B. $\frac{3x + 4y}{X}$
C. $\frac{2x + 8y}{X}$
D. $X + 8y$

32. A cylinder of height 20cm and a radius 7cm is cut into parts along the diameter. What is the volume of each part in cm^3 ? (Take $\pi=22/7$)
- 3 080
 - 440
 - 1 540
 - 385
33. Angle EFG with EF =6.2cm, angle EFG = 60° and angle FEG = 40° . Draw a perpendicular from G to meet EF at H. What is the measure of line EH?
- 5.4cm
 - 3.5cm
 - 4.1cm
 - 2.1cm
34. The length of a rectangular plot is 30m and the width is 20m. Each side of the plot is increased by 10%. What is the increase in the area of the plot?
- 6m^2
 - 126m^2
 - 600m^2
 - 726m^2
35. A rectangular tank whose base measures 1.2m by 80cm contain water to a height of 1.5m How much water, in litres is in the tank?
- 14.4
 - 144
 - 1 440
 - 14 400
36. A rectangular plot was fenced by two equal strands of wire whose total length was 440m. What was the radius of the plot? (Take $\pi =22/7$)
- $17\frac{1}{2}$
 - 35
 - 70
 - 140
37. Bongo gave 0.12 of his land to his wife, 0.25 to his son and 0.3 to his daughter. If he had 2.4 hectares of land, how many hectares was left with?
- 0.72
 - 0.288
 - 0.6
 - 0.792
38. Mwende bought the following items to make a dress:
- 2 $\frac{1}{2}$ m of dress material @ sh 275
 - 6 buttons @ sh. 7. 50
 - Three reels of threads for sh. 70
- She also paid sh. 450 for making the dress. If she had 1 500, how much money was she left with?

- A. Sh. 247. 50
- B. Sh. 1 252.50
- C. Sh. 107.50
- D. Sh. 697.50

39. The scale drawing below represents four towns U,V,W and X. The shortest distance from town V to W is 450 km.



What is the distance from town U to town X through town V and W?

- A. 100km
 - B. 1 350km
 - C. 2 480km
 - D. 1 130km
40. On a farm there are cows, goats and sheep. The number of cows while the number of cows while the number of sheep is 25 less than the number of goats. If the number of goats is g , how many animals are on the farm?
- A. $2\frac{1}{2}g - 25$
 - B. $2\frac{1}{2}g + 25$
 - C. $4g - 25$
 - D. $G - 25$
41. What is the value of y in the equation $\frac{2}{3}(6y - 2) = 2y + 4$?
- A. $2\frac{2}{3}$
 - B. 3
 - C. $\frac{8}{9}$.
 - D. $1\frac{1}{3}$
42. The area of a right-angled triangular plot 120m^2 The length of the shortest side is 10m. What is the length of the longest side?
- A. 22m
 - B. 26m
 - C. 12m
 - D. 34m
43. What is the value of $\frac{2\frac{1}{8} + \frac{1}{4}(1\frac{3}{8} - \frac{7}{8})}{3\frac{1}{8} \times \frac{3}{4} \div \frac{3}{8}}$

- A. $\frac{51}{200}$
B. $\frac{19}{100}$
C. $\frac{9}{25}$
D. $1\frac{11}{25}$
44. Juma bought a blouse and a shirt from a shop while allowed a 10% discount on the marked price of each item. He paid a total of sh. 630. If he paid sh. 270 for the blouse, what was the marked price of the shirt?
- A. Sh. 324
B. Sh. 360
C. Sh. 396
D. Sh. 400
45. John bought 50kg of sugar which he packed in packets as follows?
One quarter of sugar in 125g packets;
One half of sugar in 250g packets;
The remainder in 500g packets, altogether, of sugar did he get?
- A. 225
B. 100
C. 125
D. 2 250
46. The shared figure below was obtained from a rectangular piece of paper from which a semicircle of diameter 14cm was removed and another semicircle of diameter 7cm added.
- What is the area of the of the shared figure? (Take $\pi = \frac{22}{7}$)
- A. $229\frac{1}{4} \text{ cm}^2$
B. $306\frac{1}{4} \text{ cm}^2$
C. $152\frac{1}{4} \text{ cm}^2$
D. $133\frac{3}{4} \text{ cm}^2$
47. Murage left town Q at 8.15 a.m for town R travelling at a speed of 90km/h, Mwebi left town R at 9.00a.m for town Q travelling at a speed of 120km/h. The two meet at a place 180km away from Q. What was the distance between towns Q and R?
- A. 330km
B. 150km
C. 300km
D. 276km
48. During an election there were four candidates K,L,M and N. Candidate M received 3 421 votes which was 1 677 votes less than L receive. The total number of valid votes cast was 23 406. How many votes did K receive?
- A. 10 673
B. 12 733
C. 14 161
D. 7 319
49. December the twelfth of 1999 was Sunday. What day of the week was twelfth April 2000?

- A. Monday
- B. Tuesday
- C. Wednesday
- D. Thursday